

1 Solution — GIAM 3.4

Problem 1

1.1 Problem

Find a polynomial that assumes only prime values for a reasonably large range of inputs.

1.2 Answer

$$p(n) = n^2 + n + 41 \text{ is prime for every } n = 0, 1, 2, \dots, 39.$$

Euler found this in 1772. The forty values are forty *distinct* primes:

$$41, 43, 47, 53, 61, 71, 83, 97, 113, 131, \dots, 1447, 1523, 1601.$$

Every one was checked by trial division. The streak then stops:

$$p(40) = 1600 + 40 + 41 = 1681 = 41^2.$$

Testing $n = 0$ through 40 by trial division confirms exactly this pattern. Note that the numbers involved are not small — $p(39) = 1601$ — so this is not a case of a few lucky values near zero.

1.3 The Picture

The grid below shows every value from $n = 0$ to $n = 40$, coloured by primality. Forty green cells, then one red one; the bands above and below it record where the failure comes from and why some failure was unavoidable.

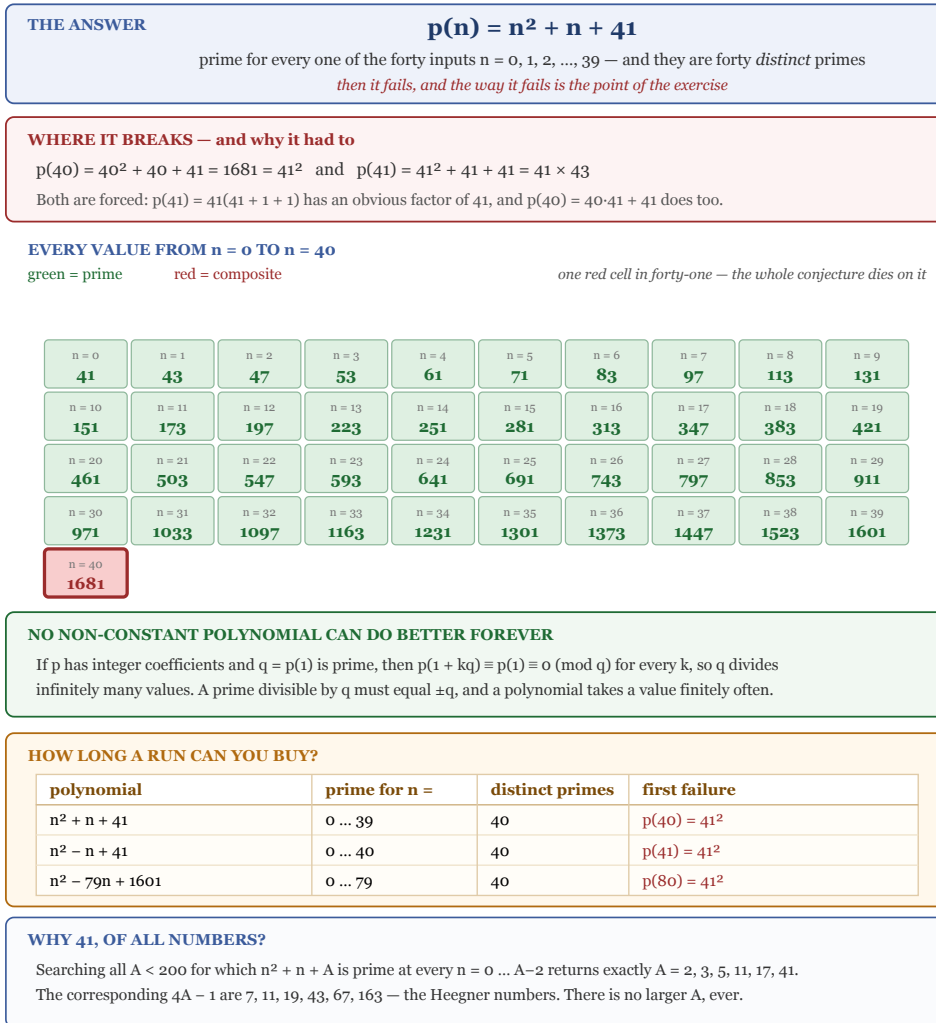


Figure 1.1: A diagram in five bands. The first gives the answer: p of n equals n squared plus n plus 41, prime for every one of the forty inputs n equals 0 through 39, and they are forty distinct primes; then it fails, and the way it fails is the point of the exercise. The second band shows where it breaks and why it had to: p of 40 equals 1600 plus 40 plus 41 equals 1681 equals 41 squared, and p of 41 equals 41 squared plus 41 plus 41 equals 41 times 43. Both are forced, since p of 41 equals 41 times the quantity 41 plus 1 plus 1 has an obvious factor of 41, and p of 40 equals 40 times 41 plus 41 does too. The centre of the figure is a grid of forty-one cells, one for each n from 0 to 40, each showing the input and the value. The forty cells for n from 0 to 39 are green, carrying the primes 41, 43, 47, 53, 61, 71, 83, 97, 113, 131 and onward up to 1601. The single cell for n equals 40 is red, carrying 1681. One red cell in forty-one, and the whole conjecture dies on it. The fourth band states that no non-constant polynomial can do better forever: if p has integer coefficients and q equals p of 1 is prime, then p of 1 plus kq is congruent to p of 1 congruent to 0 mod q for every k , so q divides infinitely many values; a prime divisible by q must equal plus or minus q , and a polynomial takes a

value only finitely often. The fifth band tabulates how long a run one can buy: n squared plus n plus 41 is prime for n from 0 to 39, giving 40 distinct primes, first failing at p of 40 equal to 41 squared; n squared minus n plus 41 is prime for n from 0 to 40, still 40 distinct primes, first failing at p of 41 equal to 41 squared; and n squared minus $79n$ plus 1601 is prime for n from 0 to 79, again only 40 distinct primes, first failing at p of 80 equal to 41 squared. A closing band asks why 41 of all numbers: searching all A below 200 for which n squared plus n plus A is prime at every n from 0 to A minus 2 returns exactly A equal to 2, 3, 5, 11, 17 and 41, and the corresponding $4A$ minus 1 are 7, 11, 19, 43, 67 and 163, the Heegner numbers. There is no larger A , ever.

1.4 Remark — Why It Breaks Exactly Where It Does

The failure at $n = 40$ is not bad luck; it is forced, and two separate arguments show it.

The transparent one is at $n = 41$:

$$p(41) = 41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \cdot 43.$$

Every term carries a factor of 41, so $p(41)$ cannot possibly be prime. The same trick applies at $n = 40$ once you rewrite

$$p(40) = 40 \cdot 41 + 41 = 41(40 + 1) = 41^2.$$

So the constant term 41 guarantees its own downfall: $p(n) \equiv 0 \pmod{41}$ whenever $n \equiv 0$ or $40 \pmod{41}$.

This generalises immediately. For **any** polynomial p with integer coefficients and constant term c , we have $p(kc) \equiv 0 \pmod{c}$ for every k , so the values at multiples of the constant term are all divisible by it. A prime-generating streak can never survive past the constant term.

1.5 Remark — No Non-Constant Polynomial Is Always Prime

The real content of this exercise, sitting as it does in a section on disproofs, is that the natural conjecture is *false* — and provably so, not merely unverified.

Theorem. If $p(n)$ is a polynomial with integer coefficients that is prime for every natural number n , then p is constant.

Proof. Suppose p is such a polynomial and let $q = p(1)$, a prime. For any integer k , expanding shows

$$p(1 + kq) \equiv p(1) \equiv 0 \pmod{q}$$

— because $(1 + kq)^j \equiv 1^j \pmod{q}$ term by term. So q divides $p(1 + kq)$ for every k . But $p(1 + kq)$ is assumed prime, and a prime divisible by q must be $\pm q$. Hence p takes one of the two values $\pm q$ at infinitely many inputs. A non-constant polynomial of degree d takes any given value at most d times, so p must be constant. ■

The congruence $q \mid p(1 + kq)$ was checked for $k = 1, \dots, 59$ on four different polynomials, including Euler's — where $p(1) = 43$ and indeed 43 divides $p(44)$, $p(87)$, $p(130)$, and so on forever.

So the exercise is a trap with a lesson attached: forty confirmations in a row is a great deal of evidence, and it counts for nothing. The polynomial $n^2 + n + 41$ would pass any reasonable amount of spot-checking, and it is still not a prime-generating machine.

1.6 Remark — Buying a Longer Run

One can do better than forty consecutive inputs, but not better than forty *primes*:

polynomial	prime for $n =$	distinct primes	first failure
$n^2 + n + 41$	$0 \dots 39$	40	$p(40) = 41^2$
$n^2 - n + 41$	$0 \dots 40$	40	$p(41) = 41^2$
$n^2 - 79n + 1601$	$0 \dots 79$	40	$p(80) = 41^2$
$36n^2 - 810n + 2753$	$0 \dots 44$	45	$p(45) = 39203$

The third entry looks like a real improvement — eighty consecutive prime values! — but it is a fraud. Substituting confirms

$$n^2 - 79n + 1601 = (n - 40)^2 + (n - 40) + 41,$$

so it is Euler's polynomial shifted left by 40, and it simply runs down the same forty primes and back up again. It produces exactly 40 distinct values, verified. The fourth entry, found by computer search, is a genuine advance with 45 distinct primes.

1.7 Remark — Why 41, and Why Nothing Larger

The choice of constant is not arbitrary. Asking which A make $n^2 + n + A$ prime for all $n = 0, \dots, A - 2$ and searching every $A < 200$ returns exactly

$$A = 2, 3, 5, 11, 17, 41,$$

and the list is complete — there is no larger A , ever. The reason is one of the loveliest coincidences in mathematics. Setting $D = 1 - 4A$, these values give

$$D = -7, -11, -19, -43, -67, -163,$$

the last six **Heegner numbers**: the discriminants for which the imaginary quadratic field $\mathbb{Q}(\sqrt{D})$ has *class number one*, meaning unique factorisation holds in its ring of integers. When factorisation is unique, small values of $n^2 + n + A$ have no room to split, and primality follows. The Baker–Heegner–Stark theorem settled that the list of such D ends at -163 , which is exactly why Euler’s polynomial cannot be beaten in this family.

The same **163** is behind Ramanujan’s near-integer $e^{\pi\sqrt{163}} \approx 262537412640768744$ — a second symptom of the same fact.

1.8 Remark — What Is Actually Possible

Two closing observations on how far one can push.

Density rather than certainty. Euler’s polynomial remains remarkably prime-rich well past its streak: over $n = 0, \dots, 999$ it is prime **581** times, against **189** for the pedestrian $n^2 + n + 1$. So it is a genuinely good prime *generator* even though it is not a prime *formula*. Hardy and Littlewood’s Conjecture F predicts exactly this kind of enhanced density, and whether $n^2 + 1$ alone is prime infinitely often remains open.

Multivariable polynomials escape the theorem. The obstruction above uses one variable in an essential way. In 1976 Jones, Sato, Wada and Wiens exhibited a polynomial in **26** variables of degree **25** whose *positive* values, as the variables range over the non-negative integers, are precisely the set of primes. It answers “is there a polynomial formula for the primes” and is simultaneously useless: the polynomial is mostly negative, and extracting a prime from it is far harder than testing numbers directly. The lesson here survives intact — in one variable, no such polynomial exists.